

P12B Macro Project: Phase 0 Summary (Evidence-First)

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1 Scope and Objective

This document summarizes what is already implemented in the Phase 0 notebook line. It reports only evidence observable in project files and executed notebook outputs.

2 Project Artifacts

Notebooks

- `/Users/stevenchung/Desktop/P12B_File/New project_/ lucas_Phase 0.ipynb`
- `/Users/stevenchung/Desktop/P12B_File/New project_/ lucas_Phase 0.5A.ipynb`
- `/Users/stevenchung/Desktop/P12B_File/New project_/ lucas_Phase 0.5B.ipynb`

Raw data

- `m2_raw.csv` (WDI, last updated date shown in file: 2025-12-19)
- `cpi_raw.csv` (WDI, last updated date shown in file: 2025-12-19)
- `gdp_raw.csv` (WDI, last updated date shown in file: 2025-12-19)
- Matching XLS downloads for the same indicators are also present.

Derived data

- `macro_growth_merged.csv`
- `final_data_for_lucas_test.csv`
- `country_diagnostics.csv`
- `lucas_panel_dataset.csv`

3 Pipeline Already Implemented

Across notebooks, the implemented pipeline is:

1. Load WDI raw files for broad money (M2), CPI, and real GDP.
2. Convert wide year columns into tidy country-year format.
3. Compute country-level log differences for M2 growth, inflation proxy, and GDP growth.
4. Merge the three growth series into a panel.
5. Run pooled OLS checks (M2 growth to inflation; M2 growth to GDP growth).
6. Build Lucas-style country averages and run cross-country OLS.
7. In 0.5B, run additional diagnostics and panel model calls (`PanelOLS`).

4 Sample Windows Used

- Phase 0: baseline pipeline without explicit hard-coded window in the same style as 0.5 notebooks.
- Phase 0.5A: forced window `YEAR_START = 1960`, `YEAR_END = 1990`.
- Phase 0.5B: forced window `YEAR_START = 1990`, `YEAR_END = 2020`.

Current exported panel (from present files) corresponds to the 0.5B style sample:

- `macro_growth_merged.csv`: 4,292 rows, 163 countries, years 1991–2020.
- `lucas_panel_dataset.csv`: same dimensions (4,292 rows, 163 countries, years 1991–2020).

5 Observed Results in Notebook Outputs

5.1 Phase 0 Notebook (reported output values)

- Pooled OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.486.
- Pooled OLS adj. R^2 (M2 $\rightarrow$ GDP growth): 0.001.
- Long-run country-average OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.884.
- Long-run country-average OLS adj. R^2 (M2 $\rightarrow$ GDP growth): -0.007.

5.2 Phase 0.5A Notebook (1960–1990 output values)

- Pooled OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.659.
- Pooled OLS adj. R^2 (M2 $\rightarrow$ GDP growth): -0.000.
- Long-run country-average OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.933.
- Long-run country-average OLS adj. R^2 (M2 $\rightarrow$ GDP growth): -0.025.

5.3 Phase 0.5B Notebook (1990–2020 output values)

- Pooled OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.525.
- Pooled OLS adj. R^2 (M2 $\rightarrow$ GDP growth): -0.000.
- Long-run country-average OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.863.
- Long-run country-average OLS adj. R^2 (M2 $\rightarrow$ GDP growth): 0.042.
- PanelOLS output appears in notebook with between- R^2 values around 0.8655 (inflation model) and -0.0500 (GDP growth model).

5.4 Recomputed from current exported panel files (cross-check)

Using current `macro_growth_merged.csv` (1991–2020):

- Corr(M2 growth, inflation): 0.7243.
- Corr(M2 growth, GDP growth): -0.0048.
- Pooled OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.5246.
- Pooled OLS adj. R^2 (M2 $\rightarrow$ GDP growth): -0.0002.
- Country-average sample with `n_obs` $\geq$ 30: 86 countries.
- Country-average OLS adj. R^2 (M2 $\rightarrow$ inflation): 0.8633.
- Country-average OLS adj. R^2 (M2 $\rightarrow$ GDP growth): 0.0423.

6 What Is Complete vs Not Complete

Complete

- Full ETL from WDI files to merged panel.
- Baseline empirical checks for Lucas-style relationship.
- Multiple window variants (0.5A, 0.5B).
- Exported datasets for downstream modeling.

Not complete

- No finalized single canonical notebook/paper workflow.
- Identification remains associational (no credible causal design yet).
- No final portfolio narrative package (problem framing, method defense, robustness protocol, final visuals).
- No explicit decision rule implemented for continue/stop based on value-of-time.

7 Main Limitation Statement

Current results are strong descriptive/associational evidence of long-run money growth and inflation co-movement, but not a causal estimate. Endogeneity, omitted variables, policy regime differences, and measurement choices are not yet addressed with a formal identification strategy.

8 Recommended Interpretation of Phase 0

Phase 0 is not “empty.” It is a functioning baseline infrastructure plus first-pass empirical signal. The gap is not data plumbing; the gap is identification quality and portfolio-grade packaging.